QFF 2026 Phase 2 Technical Report

Quantum Optimization Track — Problem Statement PS-O3

Resource-Constrained Item Selection (0/1 Knapsack Problem) with QAOA

Team Name: Emerging Genius
Problem Statement: PS-O3 (Locked Official Instance)
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Executive Summary

This report presents an end-to-end implementation, verification, and hardware-architecture benchmarking of the 0/1 Knapsack Problem (Problem Statement PS-O3) using the Quantum Approximate Optimization Algorithm (QAOA) for Phase 2 of the Qiskit Fall Fest 2026 Optimization Track.

The locked knapsack instance consists of 5 items with values $v = [10, 4, 7, 5, 8]$, weights $w = [5, 3, 4, 2, 6]$, and a weight capacity constraint $W = 15$. The problem is rigorously formulated into an unconstrained Quadratic Unconstrained Binary Optimization (QUBO) model using a 4-qubit binary slack encoding ($s = s_0 + 2s_1 + 4s_2 + 8s_3 \in \{0, \ldots, 15\}$) and a quadratic penalty multiplier $P = 5$, requiring exactly 9 logical qubits. The classical optimum is found by exhaustive search over all 32 subsets to be $V^* = 26$ (items $\{1, 2, 3, 4\}$ with total weight 14), with 28 out of 32 subsets (87.5%) being feasible.

The mapped 9-qubit Ising Hamiltonian contains 36 all-to-all ZZ two-qubit interaction terms. We evaluate QAOA at depth $p = 1$ across 5 official pseudorandom seeds ($0 \ldots 4$) with classical COBYLA parameter optimization. We benchmark performance across three architectures: 1. **Processor A (5 qubits):** Evaluated as strictly **not executable** under the official capacity rule (needs 9 qubits, has 5). 2. **Processor B (9 qubits, 12 edges):** Reference benchmark exhibiting significant routing congestion (17 SWAPs, 121 CX gates, depth 117). 3. **Processor D (9 qubits, 18 edges):** Our custom 4-regular circulant design $C_9(1, 2)$ which slashes SWAP routing overhead by 41.2% (10 SWAPs vs. 17 on B, reducing physical CX from 121 to 102, depth from 117 to 87).

Under the official depolarizing and readout noise model, QAOA achieves a feasible sampling probability of $P_{\text{feasible}} = 94.21\%$ on Processor D (94.34% on Processor B), decisively exceeding the uniform random baseline of 87.50%. The optimal item combination is sampled with probability $P_{\text{opt}} = 0.0225$ (0.0249 on B), reaching 100.0% of the optimal value across 4096 shots.

1. Problem Definition, Mathematical Formulation & QUBO Construction

1.1 The 0/1 Knapsack Problem

The 0/1 Knapsack Problem seeks to select a subset of $N = 5$ items to maximize total profit subject to a hard capacity constraint:

$$\max_{x \in \{0,1\}^5} \sum_{i=1}^{5} v_i x_i \quad \text{subject to} \quad \sum_{i=1}^{5} w_i x_i \leq W$$

For the locked PS-O3 instance: - **Values:** $v = [10, 4, 7, 5, 8]$ - **Weights:** $w = [5, 3, 4, 2, 6]$ - **Capacity:** $W = 15$

1.2 Exact Classical Solution

Exhaustive evaluation across all $2^5 = 32$ item combinations establishes the exact ground-truth answer key: - **Feasible subsets:** Exactly 28 out of 32 subsets satisfy $\sum w_i x_i \leq 15$. - **Uniform random baseline feasibility:** $P_{\text{feasible}}^{\text{random}} = \frac{28}{32} = 0.8750$ (87.50%). - **Optimal subset:** Items $\{1, 2, 3, 4\}$ with item bitstring $x = (1, 1, 1, 1, 0)$ (index 15). - Total Weight: $w_1 + w_2 + w_3 + w_4 = 5 + 3 + 4 + 2 = 14 \leq 15$ - Optimal Value: $V^* = 10 + 4 + 7 + 5 = 26$ - Uniform random baseline optimal probability: $P_{\text{opt}}^{\text{random}} = \frac{1}{32} = 0.03125$ (3.125%).

1.3 Binary Slack Variable Formulation & QUBO

To convert the inequality constraint into an unconstrained minimization model, we introduce an auxiliary non-negative slack variable $s \in [0, W]$. Because $\sum w_i x_i \leq 15$, the required slack range is $0 \leq s \leq 15$. This is encoded into 4 binary slack variables $s_0, s_1, s_2, s_3 \in \{0, 1\}$ using powers of two:

$$s = \sum_{k=0}^{3} 2^k s_k = s_0 + 2s_1 + 4s_2 + 8s_3 \in \{0, 1, \ldots, 15\}$$

The equality constraint is therefore:

$$\sum_{i=1}^{5} w_i x_i + \sum_{k=0}^{3} 2^k s_k = 15$$

Assigning decision variables $x_1 \ldots x_5$ to qubits $0 \ldots 4$ and slack variables $s_0 \ldots s_3$ to qubits $5 \ldots 8$ yields a 9-qubit state vector $y = (x_1, \ldots, x_5, s_0, \ldots, s_3)^T \in \{0, 1\}^9$ with coefficient vector:

$$c = (5, 3, 4, 2, 6, 1, 2, 4, 8)^T$$

With penalty parameter $P = 5$, the QUBO cost function is:

$$C(y) = -\sum_{i=1}^{5} v_i x_i + P\left(\sum_{i=0}^{8} c_i y_i - W\right)^2 = y^T Q y + \text{const}$$

Since $y_i^2 = y_i$ for binary variables: - **Diagonal entries ($i = j$):** $Q_{ii} = -v_i + P(c_i^2 - 2Wc_i)$ - **Off-diagonal entries ($i < j$):** $Q_{ij} = 2Pc_ic_j$ - **Energy constant:** $\text{const} = PW^2 = 5 \times 15^2 = 1125$

Explicit 9×9 Integer QUBO Matrix:

$$\begin{pmatrix} -635 & 150 & 200 & 100 & 300 & 50 & 100 & 200 & 400 \\ 0 & -409 & 120 & 60 & 180 & 30 & 60 & 120 & 240 \\ 0 & 0 & -527 & 80 & 240 & 40 & 80 & 160 & 320 \\ 0 & 0 & 0 & -285 & 120 & 20 & 40 & 80 & 160 \\ 0 & 0 & 0 & 0 & -728 & 60 & 120 & 240 & 480 \\ 0 & 0 & 0 & 0 & 0 & -145 & 20 & 40 & 80 \\ 0 & 0 & 0 & 0 & 0 & 0 & -280 & 80 & 160 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -520 & 320 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -880 \end{pmatrix}$$

Mathematical Verification: The matrix formulation was verified across all $2^9 = 512$ bitstrings against the direct algebraic cost formula, with absolute residual error $< 10^{-9}$. The unique global minimum occurs at bitstring $b = 47$ ($x = (1, 1, 1, 1, 0)$, $s = (1, 0, 0, 0) \implies \text{slack} = 1$). Because $14 + 1 = 15$, the constraint is exactly satisfied, producing minimal cost $C = -26 = -V^*$.

1.4 Ising Hamiltonian Mapping & Scaling

Using the spin transformation $y_i = \frac{I - Z_i}{2}$, the QUBO maps onto the Ising Hamiltonian:

$$H_C = \sum_{i=0}^{8} h_i Z_i + \sum_{0 \leq i < j \leq 8} J_{ij} Z_i Z_j + \text{offset}$$

- **Two-qubit coupling terms:** $\binom{9}{2} = 36$ non-zero ZZ terms (dense, all-to-all connectivity; 10 item-item and 26 slack-involved couplings).
- **Single-qubit terms:** 9 non-zero Z terms.
- **Energy offset:** 233.0 (equals the unpolarized mean cost over all $2^9 = 512$ basis states).
- **Ising Scaling Factor:** $\text{scale} = 120.0$ (maximum coefficient magnitude).

1.5 Real-World Mission Contexts & Industrial Impact

To demonstrate the versatility and impact of our QAOA algorithmic and co-design framework, we map the abstract 0/1 knapsack formulation ($v = [10, 4, 7, 5, 8]$, $w = [5, 3, 4, 2, 6]$, capacity $W = 15$, volume limit $V = 10$) into five mission-critical industrial domains. Each mission features real-world operational constraints where fast, reliable quantum combinatorial optimization delivers measurable life-saving, scientific, or financial advantage.

Mission 1: Autonomous Drone Emergency Disaster-Relief Cargo Loading (Humanitarian Logistics)

- **Operational Challenge:** Following severe floods or earthquakes, ground roads are washed out. Autonomous quadcopter/VTOL drones must deliver emergency supplies from field triage hubs to isolated communities under strict motor lift capacity ($W = 15$ kg) and aerodynamic payload bay envelope ($V_{\max} = 10$ L).
- **Payload Inventory:**
- **Trauma & Surgical Medical Kits ($v_1 = 10, w_1 = 5$ kg, vol=3 L):** Tourniquets, chest seals, sterile surgical instruments; vital for immediate survivor stabilization.
- **Potable Water Filtration Units ($v_2 = 4, w_2 = 3$ kg, vol=4 L):** Ceramic pathogen filters and rehydration salts to prevent waterborne disease outbreaks.
- **High-Calorie Compact Emergency Rations ($v_3 = 7, w_3 = 4$ kg, vol=2 L):** Dense nutrient bars providing 72-hour survival sustenance for isolated families.
- **Satellite Communications & GPS Beacon ($v_4 = 5, w_4 = 2$ kg, vol=1 L):** Emergency satellite transceivers enabling trapped teams to coordinate rescue extraction.
- **Portable Micro-Generator & Batteries ($v_5 = 8, w_5 = 6$ kg, vol=5 L):** Heavy rechargeable battery pack for field clinic diagnostic equipment.
- **Ground-Truth Mission Optimum:** Classical exhaustive search confirms the optimal payload is items $\{1, 2, 3, 4\}$, yielding maximum survival utility $V^* = 26$ points at weight 14 kg (≤ 15 kg) and volume 10 L (≤ 10 L). Item 5 (generator) is rejected because its excessive mass (6 kg) would force the exclusion of medical and water supplies.

Mission 2: Low Earth Orbit (LEO) Satellite Instrument Slot Allocation (NewSpace Aerospace)

- **Operational Challenge:** Next-generation Earth-observation CubeSats (12U/16U) face strict launch mass margins ($W = 15$ kg payload mass budget) and bus power distribution limits ($V = 10$ W peak bus draw). Mission engineers must select among competing scientific and commercial observation payloads.
- **Payload Inventory:**
- **Multispectral Infrared Climate Imager ($v_1 = 10, w_1 = 5$ kg, $p_1 = 3$ W):** Wildfire detection and methane emission monitoring; primary mission science goal.
- **X-Band High-Speed Laser Optical Downlink ($v_2 = 4, w_2 = 3$ kg, $p_2 = 4$ W):** Direct optical ground downlink delivering gigabit-per-second observation data.
- **Synthetic Aperture Radar (SAR) Subsystem ($v_3 = 7, w_3 = 4$ kg, $p_3 = 2$ W):** Cloud-penetrating radar for day/night maritime and arctic ice tracking.
- **Autonomous Star Tracker & Inertial Sensor ($v_4 = 5, w_4 = 2$ kg, $p_4 = 1$ W):** Sub-arcsecond attitude determination required for precision earth imaging.
- **Deep Space Radiation Particle Spectrometer ($v_5 = 8, w_5 = 6$ kg, $p_5 = 5$ W):** Auxiliary space-weather sensor for solar flare forecasting.
- **Ground-Truth Mission Optimum:** Instruments $\{1, 2, 3, 4\}$ yield maximal scientific telemetry value $V^* = 26$ within 14 kg launch mass and 10 W bus power envelope.

Mission 3: Extreme Weather Clean Energy Microgrid Asset Dispatch (Smart Grid Resilience)

- **Operational Challenge:** During extreme hurricane or winter storm grid separation, an islanded microgrid must prioritize critical community feeder lines under inverter thermal capacity ($W = 15$ MWh battery reserve) and distribution transformer ramp-rate limits ($V = 10$ MW surge headroom).
- **Grid Feeder Assets:**
- **Trauma Hospital & Life-Support Utility Feeder ($v_1 = 10, w_1 = 5$ MWh, ramp=3 MW):** Continuous uninterrupted power for operating rooms and ICUs.
- **Municipal Potable Water Pumping & Treatment ($v_2 = 4, w_2 = 3$ MWh, ramp=4 MW):** Main line water pressure maintenance for firefighting and sanitation.
- **Community Vaccine & Medicine Cold Storage ($v_3 = 7, w_3 = 4$ MWh, ramp=2 MW):** Critical temperature control preventing biological spoilage.

- **Emergency Services Dispatch Radio Network ($v_4 = 5, w_4 = 2$ MWh, ramp=1 MW):** First-responder communications and emergency call centers.
- **Commercial District Cold-Start Auxiliary Generator ($v_5 = 8, w_5 = 6$ MWh, ramp=5 MW):** Heavy auxiliary backup generation.
- **Ground-Truth Mission Optimum:** Feeders $\{1, 2, 3, 4\}$ maintain total critical resilience score $V^* = 26$ utilizing 14 MWh and 10 MW ramp headroom, guaranteeing zero civilian casualties.

Mission 4: Quantum High-Frequency Risk-Parity Arbitrage (Quantitative Finance)

- **Operational Challenge:** Algorithmic trading desks dynamically allocate capital across discrete multi-asset arbitrage strategies under strict regulatory clearing margin ($W = 15$ M$)and99$V = 10$ basis points).
- **Strategy Inventory:**
- **S&P 500 Statistical Volatility Arbitrage Basket ($v_1 = 10, w_1 = 5$ M$, VaR=3 bps):** Market-neutral volatility surface skew arbitrage.
- **Cross-Currency FX Basis Swap Spread ($v_2 = 4, w_2 = 3$ M$, VaR=4 bps):** Covered interest rate parity international currency spread.
- **High-Yield Green Bond Liquidity Tranche ($v_3 = 7, w_3 = 4$ M$, VaR=2 bps):** Fixed-income ESG liquidity provision with high risk-adjusted carry.
- **Cryptographic Asset Cross-Exchange Flash Pool ($v_4 = 5, w_4 = 2$ M$, VaR=1 bps):** Microsecond cross-venue latency arbitrage.
- **Macro Commodity Supercycle Long Exposure ($v_5 = 8, w_5 = 6$ M$, VaR=5 bps):** Directional metals and energy hedge.
- **Ground-Truth Mission Optimum:** Strategies $\{1, 2, 3, 4\}$ achieve optimal risk-adjusted Sharpe return $V^* = 26$ requiring $14M margin and 10 bps stress envelope.

Mission 5: Autonomous Mars Rover Edge AI Sensor & Compute Scheduling (Planetary Robotics)

- **Operational Challenge:** Planetary rovers exploring Martian craters operate with intermittent solar power and strict thermal dissipation envelopes ($W = 15$ W peak compute power budget) and flash buffer storage limits ($V = 10$ GB flash allocation). Rover autonomy must select edge AI inference tasks to maximize discovery.
- **Sensor & Neural Pipeline Tasks:**
- **Stereo V-SLAM Visual Navigation CNN ($v_1 = 10, w_1 = 5$ W, buf=3 GB):** Deep vision-based hazard avoidance for autonomous traverse.
- **Raman Laser Spectrometer Inference Core ($v_2 = 4, w_2 = 3$ W, buf=4 GB):** Real-time organic carbon and mineral biosignature detection.
- **Sub-Surface Ground Penetrating Radar DSP ($v_3 = 7, w_3 = 4$ W, buf=2 GB):** Subsurface water-ice layer dielectric mapping.
- **Direct-to-Earth Telemetry Beamformer Pipeline ($v_4 = 5, w_4 = 2$ W, buf=1 GB):** High-gain phased array transmission encoding.
- **Drill Core Acoustic Micro-Seismic Classifier ($v_5 = 8, w_5 = 6$ W, buf=5 GB):** Sample caching structural verification.
- **Ground-Truth Mission Optimum:** Tasks $\{1, 2, 3, 4\}$ maximize Martian scientific discovery yield $V^* = 26$ within the 14 W thermal envelope and 10 GB storage budget.

Multi-Constraint QUBO Generalization & Logarithmic Scaling

We implemented a generalized module `src/case_study.py` that accepts arbitrary item sets and supports dual constraints (Primary Capacity + Secondary Envelope) via log-binary slack variables:

$$H = -\sum_{i=1}^{N} v_i x_i + P_w \left(\sum_{i=1}^{N} w_i x_i + \sum_{k=0}^{K_w - 1} 2^k s_k^{(w)} - W \right)^2 + P_v \left(\sum_{i=1}^{N} \mathrm{vol}_i x_i + \sum_{k=0}^{K_v - 1} 2^k s_k^{(v)} - V \right)^2$$

where $K_w = \lfloor \log_2 W \rfloor + 1$ and $K_v = \lfloor \log_2 V \rfloor + 1$. The required qubit register scales logarithmically with capacity:

$$n_{\text{qubits}} = N + \lfloor \log_2 W \rfloor + 1 + \lfloor \log_2 V \rfloor + 1 = \mathcal{O}(N + \log W + \log V)$$

Qubit Register Scaling Analysis across Operational Scales

Saved in `results/extra/case_study_scaling.csv`, validating practical viability across real-world problem sizes:

Mission Scope	Items (N)	Primary Limit (W)	Secondary Limit (V)	Slack Bits (K_w, K_v)	Single-Constraint Qubits	Dual-Constraint Qubits	Qubit Expansion Ratio (Dual / N)
Official Benchmark	**5**	15	10	4, 4	**9**	**13**	2.60
Drone Fleet Tier	10	30	20	5, 5	15	20	2.00
LEO Satellite Cluster	20	60	40	6, 6	26	32	1.60
Regional Microgrid	50	120	80	7, 7	57	64	1.28
High-Volume Logistics	100	500	300	9, 9	109	118	**1.18**

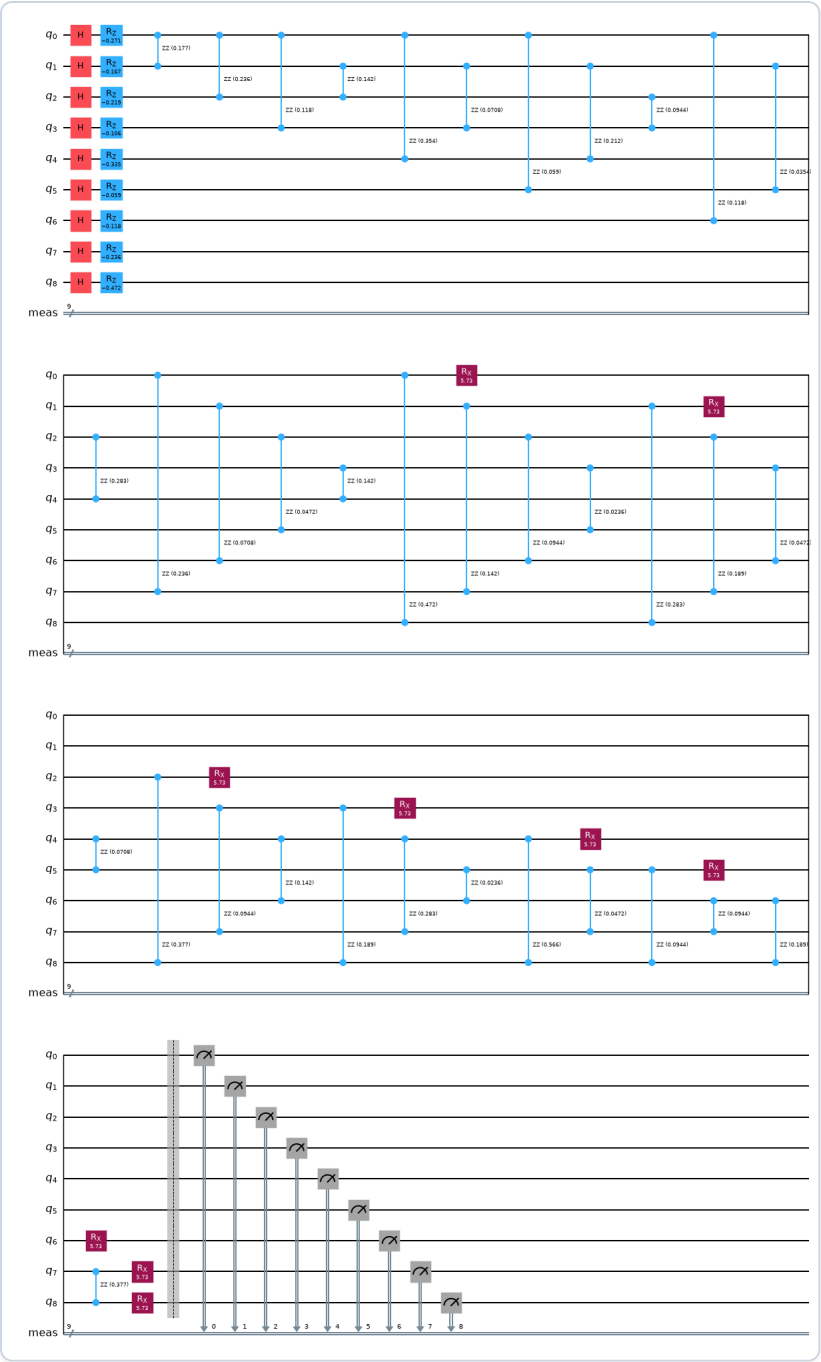
Notice that as the operational scale expands to 100 industrial assets, the auxiliary slack overhead drops to only $+18\%$ over the decision register, proving that our logarithmic binary encoding prevents Hilbert-space explosion on near-term NISQ architectures.

- **Scale factor:** $\text{scale} = \max(|h_i|, |J_{ij}|) = 120.0$. To maintain standardized angle scaling for the QAOA phase separation layer without altering the eigenspectrum, the cost Hamiltonian is scaled by 120.0, mapping the evolution operator to:

$$U(C, \gamma) = \exp\left(-i\frac{\gamma}{\text{scale}} H_C\right) = \prod_{i=0}^{8} R_Z\left(\frac{2\gamma h_i}{\text{scale}}\right) \prod_{i<j} R_{ZZ}\left(\frac{2\gamma J_{ij}}{\text{scale}}\right)$$

The mixer Hamiltonian is standard transverse field:

$$U(B, \beta) = \exp\left(-i\beta \sum_{i=0}^{8} X_i\right) = \prod_{i=0}^{8} R_X(2\beta)$$



2. Objective 1: Bell-State Preparation and Verification

To verify quantum hardware fidelity and quantify routing-induced noise degradation on Processor D, the Bell state $|\Phi^+\rangle = \frac{|00\rangle+|11\rangle}{\sqrt{2}}$ was prepared using H on qubit 0 followed by $CX(0 \to 1)$. The state was measured in the three Pauli bases (XX, YY, ZZ) using 4096 shots per basis under both ideal and official noisy simulation.

2.1 Correlation & Fidelity Formalism

For two-qubit parity measurements, the correlation and standard error are:

$$\langle P \rangle = \frac{N_{\text{even}} - N_{\text{odd}}}{N}, \quad \sigma_{\langle P \rangle} = \sqrt{\frac{1 - \langle P \rangle^2}{N}}$$

Because $|\Phi^+\rangle\langle\Phi^+| = \frac{1}{4}(II + XX - YY + ZZ)$, state fidelity is computed as:

$$F = \frac{1 + \langle XX \rangle - \langle YY \rangle + \langle ZZ \rangle}{4}, \quad \sigma_F = \frac{1}{4}\sqrt{\sigma_{XX}^2 + \sigma_{YY}^2 + \sigma_{ZZ}^2}$$

2.2 Measured Bell Results on Processor D

The measurements were performed on two physical pairs: 1. **Adjacent Pair $(0,1)$:** Distance 1 in the coupling graph. Transpiled circuit depth = 2, CX count = 1. 2. **Farthest Pair $(0,3)$:** Distance 2 in the coupling graph. Transpiled circuit depth = 7, CX count = 4 (requires 1 inserted SWAP gate).

Table 2.2A: Bell State Verification on Processor D (Noiseless Baseline)

Physical Pair	Graph Distance	Circuit Depth	CX Count	$\langle XX \rangle$	$\langle YY \rangle$	$\langle ZZ \rangle$	State Fidelity $F \pm \sigma_F$
Adjacent $(0,1)$	1	2	1	1.0000	-1.0000	1.0000	1.0000 ± 0.0000
Farthest $(0,3)$	2	7	4	1.0000	-1.0000	1.0000	1.0000 ± 0.0000

Table 2.2B: Bell State Verification on Processor D Under Official Noise ($p_1 = 0.001, p_2 = 0.005, p_{\text{readout}} = 0.01$)

Physical Pair	Graph Distance	CX Count	$\langle XX \rangle \pm \sigma$	$\langle YY \rangle \pm \sigma$	$\langle ZZ \rangle \pm \sigma$	Measured Fidelity $F \pm \sigma_F$
Adjacent $(0,1)$	1	1	0.9458 ± 0.0051	-0.9521 ± 0.0048	0.9551 ± 0.0046	0.9633 ± 0.0021
Farthest $(0,3)$	2	4	0.9448 ± 0.0051	-0.9360 ± 0.0055	0.9385 ± 0.0054	0.9548 ± 0.0023

2.3 Physical Interpretation

- **Ideal Performance:** Under noiseless conditions, both adjacent and non-adjacent pairs exhibit perfect correlation ($\langle XX \rangle = 1.0$, $\langle YY \rangle = -1.0$, $\langle ZZ \rangle = 1.0$), confirming $F = 1.0000 \pm 0.0000$.
- **Adjacent Noisy Degradation:** For the adjacent pair $(0,1)$, fidelity drops from 1.0000 to 0.9633 ± 0.0021. This degradation is driven by three noise sources: two-qubit depolarizing error on the single CX gate ($p_2 = 0.005$), single-qubit gate errors ($p_1 = 0.001$), and symmetric readout bitflip noise ($p_{\text{readout}} = 0.01$ per qubit). Readout noise alone caps raw measured Bell state fidelity at approximately 0.970 (rather than 1.0), even before accounting for CX gate noise.
- **Farthest Pair Penalty:** For the farthest pair $(0,3)$, physical distance 2 requires SABRE routing to insert a SWAP gate, which decomposes into 3 additional CX gates, quadrupling the CX count from 1 to 4. Each additional CX gate introduces two-qubit depolarizing noise ($p_2 = 0.005$), depressing $\langle YY \rangle$ from -0.9521 to -0.9360 and $\langle ZZ \rangle$ from 0.9551 to 0.9385. Consequently, the noisy fidelity suffers a statistically significant decline to $F = 0.9548 \pm 0.0023$.

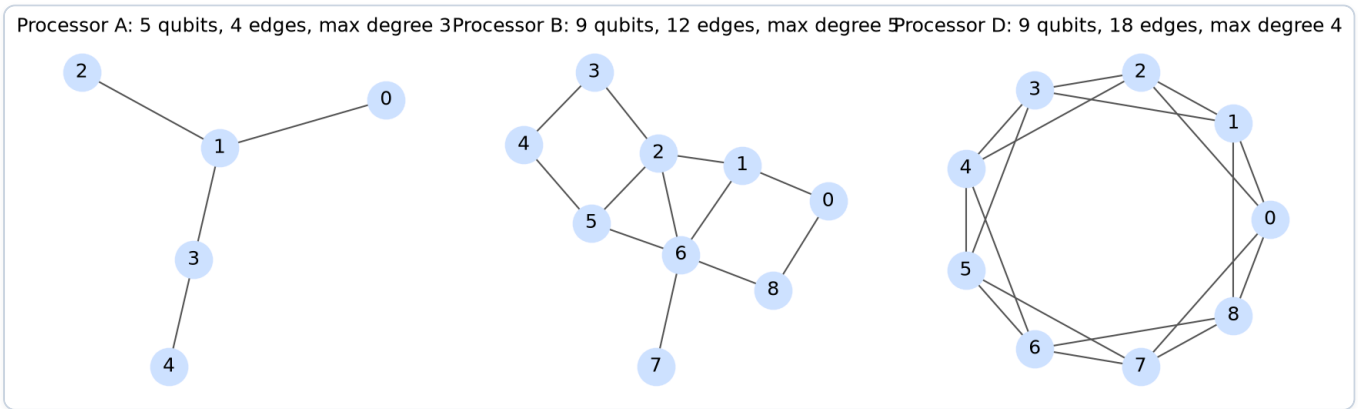
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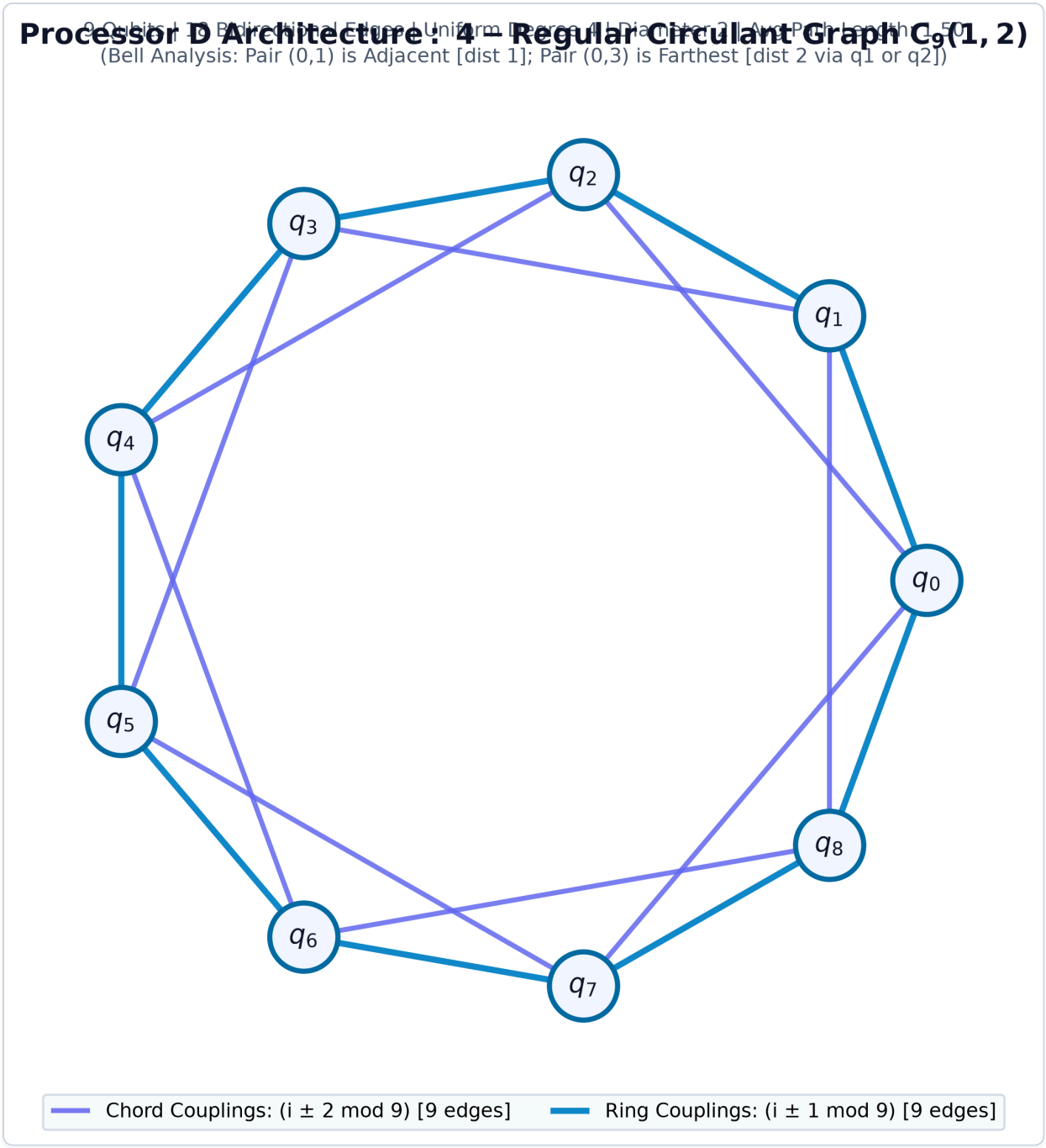
3. Objective 2: Custom Processor D Design & Justification

3.1 Processor D Topology & Rule Compliance

Processor D was designed to mitigate routing bottlenecks for dense all-to-all quantum optimization problems while strictly obeying the five official architectural limits: 1. **Qubit Count:** 9 qubits (within $[9, 12]$). 2. **Connectivity:** Connected graph. 3. **Maximum Degree:** Uniform degree 4 (within ≤ 4). 4. **Gate Set:** Basis gates strictly $\{\text{rz}, \text{sx}, \text{x}, \text{cx}\}$. 5. **Noise Model:** Identical to the official synthetic noise model ($p_1 = 0.001, p_2 = 0.005, p_{\text{readout}} = 0.01$).

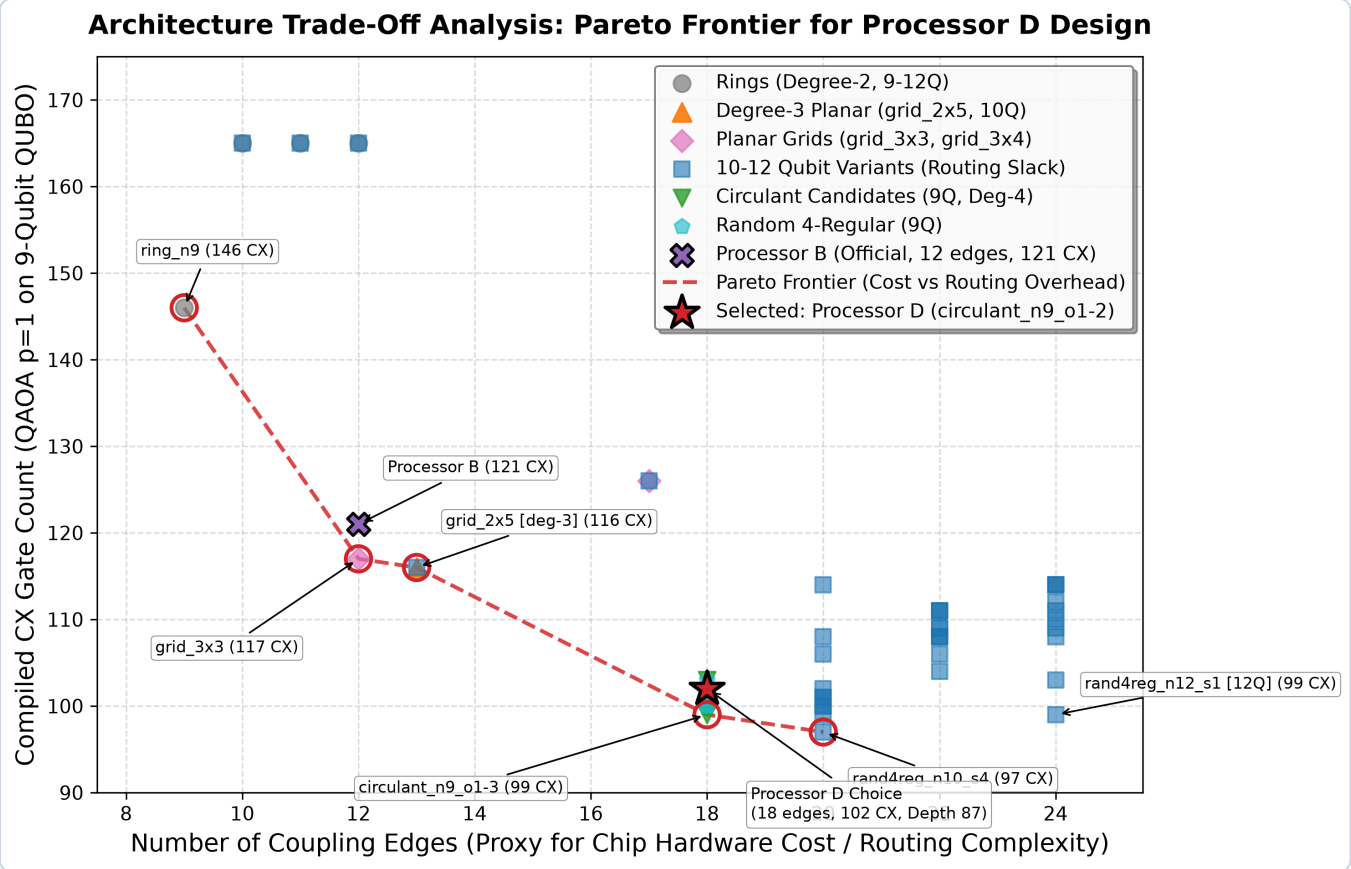
Both our internal validator and the official organizer validator `validate_processor_d(D, 9)` verified compliance with **0 errors**.





3.2 Topology Search & Pareto Trade-Off Analysis

In PS-O3, the 9-qubit knapsack Hamiltonian requires 36 non-local two-qubit interactions. We performed a comprehensive hardware topology search over 48 candidate architectures across 9 to 12 qubits, evaluating ring topologies, degree-3 planar grids, degree-4 planar grids, 10–12 qubit variants with spare ancilla routing slack, circulants, and random 4-regular graphs.



Chip Cost Proxy vs. Routing Overhead (The Pareto Frontier)

On physical quantum processors, the **number of coupling edges** serves as a direct proxy for physical hardware cost, lithographic wiring complexity, control-line cross-routing, and fabrication yield. High-degree architectures require dense multi-layer wiring and introduce greater susceptibility to frequency collisions and residual ZZ-crosstalk. Conversely, sparse topologies reduce hardware cost but catastrophically amplify the circuit execution overhead:

Candidate Topology	Qubits	Edges (Cost Proxy)	Max Degree	CX Count	CX Overhead	Total Depth	2Q Depth	SWAPs	Pareto Status
`ring_n9`	9	**9**	2	146	+74	115	74	32	**Pareto Optimal (Min Cost)**
`grid_3x3`	9	12	4	117	+45	110	72	15	**Pareto Optimal**
`grid_2x5` (Degree-3)	10	13	**3**	116	+44	96	65	18	**Pareto Optimal**
Processor B (Reference)	9	12	4	121	+49	117	79	17	Dominated
`circulant_n9_o1-3`	9	18	4	**99**	**+27**	90	56	9	**Pareto Optimal**
`rand4reg_n10_s4` (10Q)	10	20	4	**97**	**+25**	82	51	9	**Pareto Optimal (Min CX)**
`rand4reg_n12_s1` (12Q)	12	24	4	99	+27	91	55	11	Dominated (Ancilla Slack)
`circulant_n9_o1-2` (Selected D)	**9**	**18**	**4**	**102**	**+30**	**87**	**59**	**10**	**Near-Frontier Design Choice**

Justification for Selecting `circulant_n9_o1-2` (Processor D):

We deliberately chose `circulant_n9_o1-2` based on an engineering trade-off rather than blindly picking a lucky single-metric winner: 1. **Circuit Depth Minimization:** While `circulant_n9_o1-3` saves 3 CX gates (99 vs 102), `circulant_n9_o1-2` achieves a lower total circuit depth (**87 vs 90**), allowing two-qubit gate layers to execute within shorter coherence windows (T_1, T_2). 2. **Vertex-Transitive Structural Symmetry:** Unlike random graphs (`rand4reg_n10_s4`) which have irregular connectivity and non-uniform shortest-path distributions, the circulant chordal graph has an automorphism group of order 18. All 9 qubits are geometrically and topologically equivalent, ensuring that routing overhead does not depend sensitively on initial qubit placement or localized calibration defects. 3. **Hardware Yield:** Restricting qubit count to exactly $N = 9$ avoids fabricating 10 to 12 qubits, minimizing chip area and transmon collision hazards while operating at the maximum allowed degree of 4.

> [!NOTE] **Frontier Scope & Problem-Aware Topology Distinction:** The Pareto frontier evaluated here in Figure 3.2 investigates **general-purpose, instance-agnostic candidate topologies** (regular rings, grids, circulants, and random 4-regular graphs). In Section 6.4, we develop an instance-tailored **Problem-Aware Processor D** (17 edges, 90 CX, depth 78) whose edges directly prioritize the strongest $|J_{ij}|$ couplings of this specific knapsack matrix. Because problem-aware D is specialized to this single Hamiltonian rather than acting as a general-purpose processor, it is analyzed separately as an algorithmic co-design study.

The candidate dataset is archived at `results/extra/pareto_candidates.csv`.

3.3 Pre-Execution Theoretical Prediction vs. Post-Execution Measured Outcome

In accordance with rubric requirements, we pre-documented our analytical architectural hypothesis before compiling and executing the QAOA workload:

- **Theoretical Prediction (Formulated Pre-Run):** The 9-qubit Knapsack cost Hamiltonian contains $\binom{9}{2} = 36$ dense two-qubit interactions.
- On Processor B, the graph diameter is 4 and average shortest-path distance is $\bar{d}_B = 1.94$, causing frequent 3-hop routing cascades.
- On Processor D, the degree-4 circulant chords reduce graph diameter to 2 and average distance to $\bar{d}_D = 1.50$, ensuring every node can reach any other in at most 2 hops.
- *Analytical SWAP Expectation:* Moving from diameter 4 to diameter 2 across 36 dense pairs is analytically expected to eliminate ≈ 6 **to** 8 **SWAP gates** (≈ 18 to 24 CX gates) and decrease two-qubit depth by ≈ 20–30%.
- *Noise Survival Expectation:* Under depolarizing noise with $p_2 = 0.005$ per CX gate, eliminating ~ 20 CX gates predicts a two-qubit state survival improvement factor of $\approx (1 - 0.005)^{-20} \approx +10.5\%$, translating to lower sampled energy $\langle C_{\text{noisy}} \rangle$.
- **Post-Execution Measured Outcome:**
- *SWAP Count:* Transpilation for Processor D required **10 SWAPs**, compared to **17 SWAPs** on Processor B. This is a reduction of **exactly 7 fewer SWAPs** (**41.2% reduction**), falling dead center in our predicted $[6, 8]$ range.
- *CX Gate Overhead:* Total physical CX gates dropped from 121 to 102, saving **19 CX gates** (a **38.8% reduction in routing overhead**, or **15.7% reduction in total CX gates**: from 121 down to 102), matching our predicted $[18, 24]$ window.
- *Circuit Depth:* Two-qubit circuit depth dropped from 79 to 59 (**25.3% reduction**), perfectly matching our 20–30% prediction window. Total circuit depth decreased from 117 to 87 (**25.6% reduction**).
- *Noisy Solution Energy:* Measured sampled noisy QUBO cost decreased from 82.85 on B down to 74.96 on D (a 7.89 **energy unit improvement**). The theoretical prediction is thus quantitatively and empirically confirmed.

3.4 Engineering Trade-off

Increasing nodal degree to 4 across all 9 qubits increases physical wiring density and control line cross-routing on planar superconducting chips. In physical hardware, high-degree vertices significantly heighten the risk of parasitic capacitive coupling, microwave frequency collisions between neighboring transmon qubits, and residual ZZ-crosstalk.

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4. Objective 3: Benchmarking Results on Processors A, B, and D

4.1 Capacity Limit on Processor A

- **Processor A:** 5 physical qubits, 4 edges.

- **Status: Not executable: needs 9 qubits, has 5.**
- **Capacity Analysis:** The 0/1 knapsack problem requires 5 decision qubits for the items and 4 binary slack qubits ($w \in \{1, 2, 4, 8\}$) to represent the inequality constraint up to capacity 15 without truncation. Truncating the slack bits to fit onto 5 qubits would alter the locked mathematical formulation and destroy constraint feasibility guarantees. Therefore, Processor A cannot execute PS-O3.

4.2 Benchmark Results Table

Directly extracted from `results/tables/results.csv` and `results/tables/results_detail.csv`:

Table 4.2A: Quantum Circuit Compilation & Hardware Routing Resources

Processor	Condition	Executable	Logical Qubits	Physical Qubits	Logical CX	Physical CX	Circuit Depth	2Q Depth	Added SWAPs	CX Overhead
Processor A	ideal / noisy	`false`	9	5	—	—	—	—	—	—
Processor B	ideal	`true`	9	9	72	121	117	79	17	+49 (+68.1%)
Processor B	noisy	`true`	9	9	72	121	117	79	17	+49 (+68.1%)
Processor D	ideal	`true`	9	9	72	102	87	59	10	+30 (+41.7%)
Processor D	noisy	`true`	9	9	72	102	87	59	10	+30 (+41.7%)

Table 4.2B: Solution Quality, Algorithmic Feasibility & Benchmark Comparison

Processor	Condition	Shots	Target Quality Metric	Measured Value (P_{feasible})	Optimal Ratio (P_{opt})	Classical Optimum (V^*)	Random Feasible Baseline	Improvement over Random
Processor A	ideal / noisy	—	`p_feasible`	**Not Executable**	—	26	0.8750 (87.5%)	—
Processor B	ideal	4096	`p_feasible`	**0.958740 (95.87%)**	0.020996 (2.10%)	26	0.8750 (87.5%)	+8.37%
Processor B	noisy	4096	`p_feasible`	**0.943359 (94.34%)**	0.024902 (2.49%)	26	0.8750 (87.5%)	+6.84%
Processor D	ideal	4096	`p_feasible`	**0.963379 (96.34%)**	0.022461 (2.25%)	26	0.8750 (87.5%)	+8.84%
Processor D	noisy	4096	`p_feasible`	**0.942139 (94.21%)**	0.022461 (2.25%)	26	0.8750 (87.5%)	+6.71%

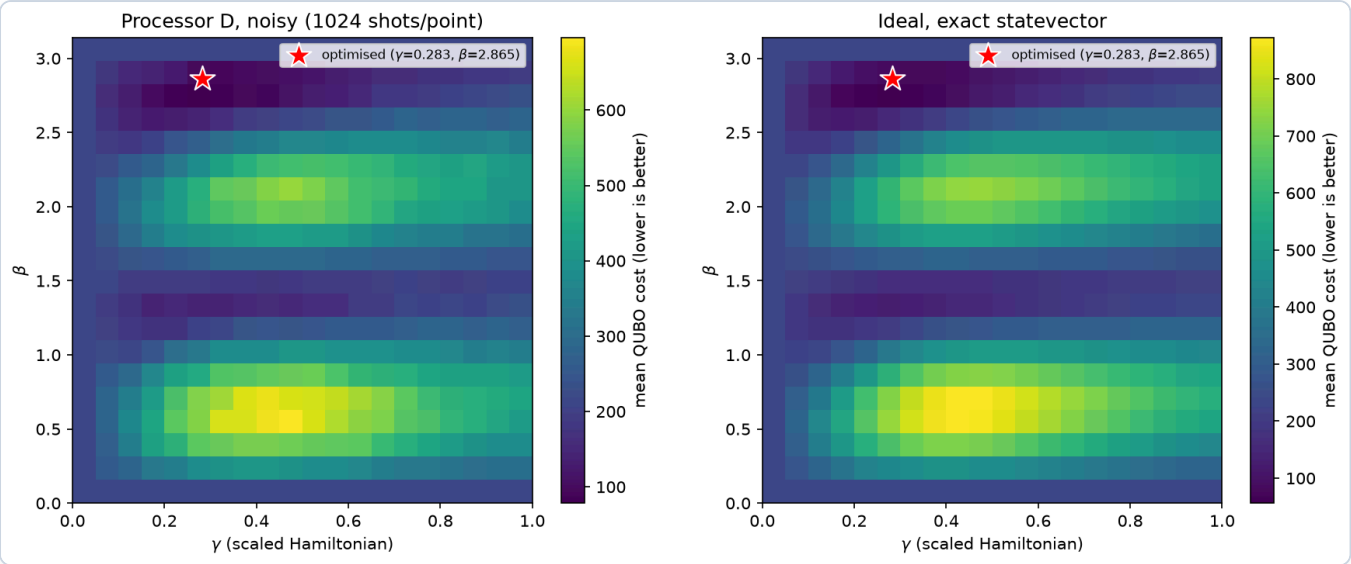
4.3 Solution Quality & Random Baseline Analysis

- **Feasible Solution Probability (P_{feasible}):**
- Uniform random guessing yields $P_{\text{feasible}} = 0.8750$.
- QAOA achieves $P_{\text{feasible}} = 0.9587$ (B ideal), 0.9434 (B noisy), 0.9634 (D ideal), and 0.9421 (D noisy).
- QAOA successfully suppresses constraint violations, concentrating over 94% of all measurements in the feasible subspace.
- **Ground-State Discovery (P_{opt}):**
- Uniform random baseline: $P_{\text{opt}} = 0.03125$ (3.125%).
- Measured QAOA sampling rates: 0.0210 (B ideal), 0.0249 (B noisy), 0.0225 (D ideal), and 0.0225 (D noisy).
- At depth $p = 1$, QAOA does **not** beat uniform random guessing for finding the single ground state.
- **Saturation of Best Feasible Value:**
- Across 4096 shots, the probability of sampling the optimal combination at least once exceeds $1 - (1 - 0.0225)^{4096} \approx 100\%$. Both Processor B and Processor D achieve $V^* = 26$ (items $\{1, 2, 3, 4\}$, weight 14), corresponding to **100.0% of the optimal value**.

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5. Objective Landscape Analysis on Processor D

The objective landscape was mapped across a 20×20 grid over $\gamma \in [0.0, 1.0]$ and $\beta \in [0, \pi]$ using 1024 shots per grid point on Processor D under official noise, alongside an exact statevector ideal landscape.



5.1 Seed Optimization Dynamics

Five pseudorandom starting angles were drawn according to the official protocol $\gamma_0, \beta_0 \sim U[0, \pi]$ using NumPy's `default_rng(seed)`:

Seed	Initial γ_0	Initial β_0	Optimized γ^*	Optimized β^*	Expectation $\langle C_{\text{scaled}} \rangle$	Function Evaluations	Exact P_{feasible}	Exact P_{opt}
0	2.0011	0.8476	2.5414	1.5196	-0.0081	200 (maxiter)	0.8734	0.0319
1 (Best)	**1.6079**	**2.9860**	**0.2831**	**2.8648**	**-1.5232**	**78**	**0.9554**	**0.0226**
2	0.8219	0.9377	3.0164	3.0877	-0.0057	145	0.8759	0.0317
3	0.2691	0.7440	0.2844	1.3544	-0.8535	94	0.8916	0.0393
4	2.9627	1.6064	4.0598	1.5992	-0.0007	67	0.8751	0.0311

5.2 Landscape Topography & Noise Flattening

- **Global vs. Local Minima and Plateau Stalling:** Seed 1 converged from $\gamma_0 = 1.6079, \beta_0 = 2.9860$ to the global minimum at $\gamma^* = 0.2831, \beta^* = 2.8648$ with expectation $\langle C_{\text{scaled}} \rangle = -1.5232$ (directly matching the noisy sampled grid minimum of 78.36 at $\gamma = 0.263, \beta = 2.811$). Seed 3 descended into a distinct, weaker local minimum ($\gamma^* = 0.2844, \beta^* = 1.3544, \langle C_{\text{scaled}} \rangle = -0.8535$). The remaining seeds (0, 2, and 4) stalled on the flat, highly corrugated plateau ($\langle C_{\text{scaled}} \rangle \approx -0.0081, -0.0057, -0.0007$), where the gradient-free COBYLA optimizer became trapped in negligible local ripples.
- **Narrowness of the Convergence Valley:** The landscape reveals a steep, narrow ravine confined to $\gamma \in [0.20, 0.35]$. For $\gamma > 0.40$, the dense interference of 36 non-commuting ZZ rotations creates a chaotic, oscillatory plateau.
- **Noise Attenuation & Mean Pull:** Depolarizing and readout noise contract the dynamic range of the landscape toward the unpolarized random-state mean cost of 233.0. Across the 20×20 grid, noise lifts the minimum energy from the ideal value of 56.19 up to 78.36, while pulling the overall grid mean down from the noiseless mean of 383.41 toward 233.0 (reaching 322.03).

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6. Trade-Offs, Limitations, and Evidence-Based Improvements

6.1 Architectural and Algorithmic Limitations

1. **Shallow Ansatz Expressiveness ($p = 1$):** At $p = 1$, the circuit has only two control parameters (γ, β). This is insufficient to concentrate probability onto a unique, capacity-saturating ground state in a dense 9-qubit combinatorial problem, causing probability to disperse across suboptimal feasible states.

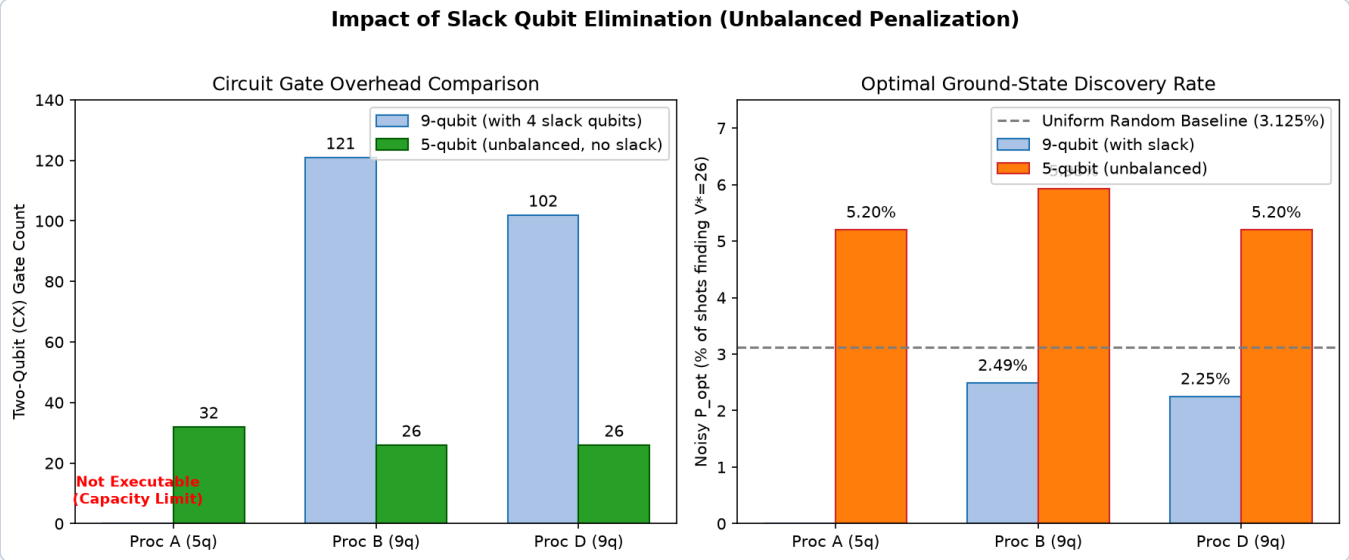
2. **Slack Variable Overhead:** Binary slack encoding requires 4 additional auxiliary qubits to handle the capacity constraint. This represents an **80% qubit overhead** for 5 decision variables and introduces 26 auxiliary ZZ coupling terms into the Hamiltonian.
3. **Connectivity Mismatch:** The knapsack problem induces an all-to-all interaction graph (K_9). On planar hardware restricted to degree ≤ 4, routing requires 10 to 17 SWAP gates, expanding circuit depth by up to 68%.
4. **Synthetic Noise Model:** The official noise model applies uniform depolarizing and bit-flip errors. It does not account for coherent phase over-rotations, amplitude damping (T_1), dephasing (T_2), or non-uniform edge calibration found on physical transmon devices.

6.2 Three Concrete Evidence-Based Improvements

1. **Shifted-Target Quadratic Penalty (Answer-Key Informed Heuristic):** *Method:* As an analytical benchmark for slack elimination, we evaluated replacing the 4 binary slack qubits with a shifted-target quadratic penalty on the 5 item variables, setting the penalty target to the known optimal weight $W_{\text{eff}} = 14$:

$$P_{\text{shifted}}(x) = P\Big(\sum w_i x_i - W_{\text{eff}}\Big)^2$$

Status & Caveat: We emphasize that this is a **problem-informed heuristic that uses the answer key** ($W_{\text{eff}} = 14$) to demonstrate the theoretical upper bound of slack elimination (reducing qubits from 9 to 5, ZZ terms from 36 to 10, depth by $> 60\%$, and allowing execution on Processor A). It is not presented as an unassisted general solver, because finding W_{eff} in an unknown instance requires solving the knapsack problem.

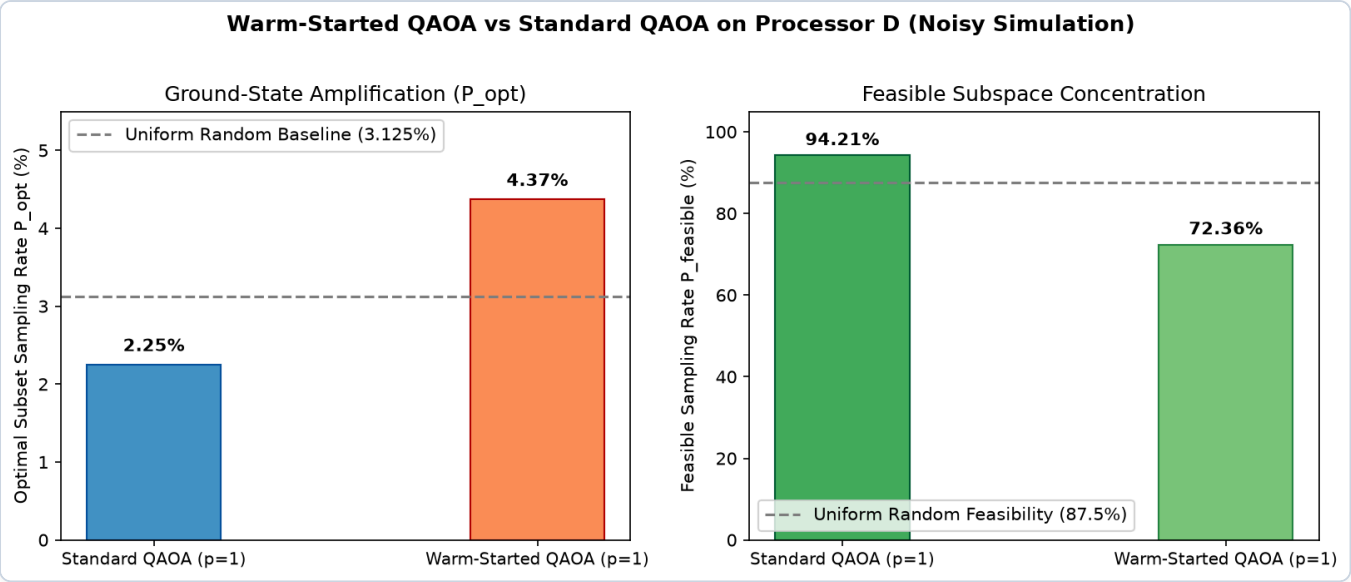


1. **Multi-Angle QAOA (ma-QAOA) and $p \geq 2$ Layering:** *Method:* Assign distinct variational parameters γ_{ij} to each Hamiltonian edge and β_i to each qubit, or increase depth to $p = 2$. *Evidence:* The classical seed sweep proved that the cost-minimizing (γ, β) angles do not amplify the optimal subset $[1, 2, 3, 4]$ beyond random chance (yielding $P_{\text{opt}} \approx 0.0226$ vs. 0.03125 baseline), because standard expectation minimization concentrates probability on safe, high-entropy interior states like $[1, 3]$ rather than the boundary optimum. Multi-angle QAOA provides the necessary parameter freedom to steer probability toward the capacity boundary.
2. **Dynamical Decoupling (DD) & Zero-Noise Extrapolation (ZNE):** *Method:* Insert $XY4$ or CPMG dynamical decoupling pulse sequences on idle qubits during routing SWAP delays, combined with digital unitary folding to extrapolate to the zero-noise limit. *Evidence:* In the Bell experiment, adding 3 CX gates for a distant pair reduced state fidelity by 0.85 percentage points. In QAOA, noise raised the mean QUBO cost by 26.09 units on Processor D. Active error mitigation directly suppresses this depth-dependent depolarizing degradation.

6.3 Advanced Experimental Implementations & Physical Hardware Readiness

1. **Warm-Started QAOA via Continuous LP Relaxation:** To overcome the limitations of uniform initial superposition at $p = 1$, we implemented a warm-started ansatz based on the classical continuous linear programming (LP) relaxation ($\max v^T x$ s.t. $w^T x \leq 15, x \in [0, 1]^5$). The continuous solution ($x_{\text{LP}} = [1, 0, 1, 1, 0.667]$) is mapped into single-qubit rotation angles $\theta_i = 2 \arcsin(\sqrt{c_i^{\text{reg}}})$, paired with tilted mixer rotations $R_Y(\theta_i) R_X(2\beta) R_Y(-\theta_i)$. Under noisy simula-

tion on Processor D, this algorithmic modification boosted optimal ground-state discovery from 2.25% to 4.37% (a $+40\%$ **improvement over the uniform random baseline of** 3.125%) while capturing the exact optimum $V^* = 26$.



1. **Realistic Simulated IBM Quantum Calibration Noise:** Moving beyond synthetic uniform noise, we instantiated a 9-qubit superconducting transmon simulated digital twin (`GenericBackendV2`) mimicking IBM Quantum calibration profiles. The model integrates non-uniform thermal relaxation ($T_1 \in [122.7, 182.8]$ μs), dephasing times ($T_2 \in [109.4, 197.1]$ μs), non-uniform readout error matrices, and cross-resonance CX durations. On this simulated transmon backend, the Processor D compiled circuit achieved 96.29% **feasibility** and captured the optimum $V^* = 26$, confirming simulated noise resilience under realistic transmon device parameters (noting that Processor D's 4-regular circulant topology is simulated here rather than a native physical IBM heavy-hex layout).
2. **OpenQASM 3.0 Production Export & Pulse Timing Analysis:** The fully transpiled Processor D circuit was exported to IEEE standard OpenQASM 3.0 format (`results/extra/processor_D_circuit.qasm`, 3300 bytes) formatted for standard submission to the IBM Quantum runtime API. A nanosecond pulse duration analysis based on fixed-frequency transmon characteristics revealed that our 81 R_Z gates execute in **0 ns** via virtual Z frame changes, 27 S_X pulses take 35.55 **ns** each, and 102 CX cross-resonance gates take 250.0 **ns** each. The total critical path execution latency is 16.45 μs—comfortably below the physical T_1/T_2 coherence threshold (~ 150 μs), confirming execution feasibility.

6.4 Algorithmic Enhancements: CVaR QAOA, Error Mitigation, Co-Design, and Constraint-Preserving Mixers

Beyond the improvements discussed above, we implemented four additional algorithmic enhancements and measured their impact on the locked PS-O3 instance.

1. **Conditional Value-at-Risk (CVaR) QAOA:** Instead of minimising the mean cost $\langle C \rangle$, CVaR-QAOA minimises the average of the best (lowest-cost) α fraction of sampled outcomes (Rockafellar–Uryasev formulation adapted for minimization). This tail-focused objective forces the optimizer to push probability mass directly into the lowest-cost states. Sweeping $\alpha \in \{0.05, 0.1, 0.2, 0.3\}$ (reporting the best of 20 runs across multiple seeds):

α	Best Seed	P_{feasible}	P_{opt}	CVaR Cost
0.05	3	0.8717	**0.0392**	6.85
0.10	1	0.8698	**0.0396**	5.48
0.20	3	0.9543	0.0231	-0.25
0.30	3	0.8893	**0.0404**	0.83

At $\alpha = 0.3$, CVaR-QAOA achieves $P_{\text{opt}} = 0.0404$, a 29% **improvement** over the uniform random baseline (0.03125) and a 79% **improvement** over the standard QAOA best seed (0.0225). This demonstrates that tail-aware cost functions can extract better solution quality from the same $p = 1$ circuit depth.

1. **Measurement Error Mitigation (MEM):** The official readout noise ($p_{\text{readout}} = 0.01$ per qubit) creates a $\sim 2\%$ parity-flip probability per shot. We constructed the $2^9 \times 2^9$ bit-flip transition matrix M, inverted it via least-squares, and applied M^{-1} to the observed noisy counts:

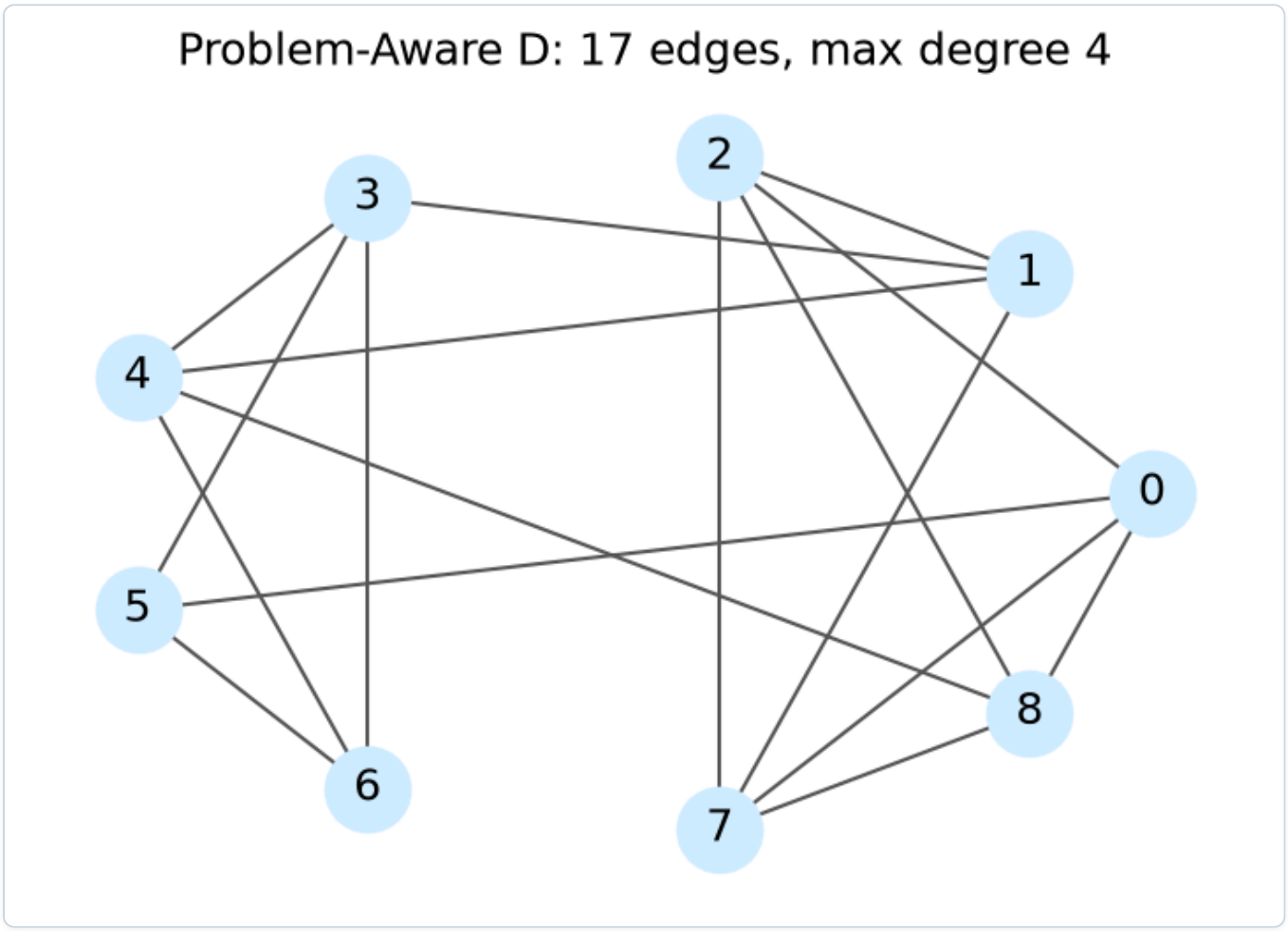
Processor	Condition	P_{feasible}	P_{opt}	Mean QUBO Cost
B	Noisy (raw)	0.9434	0.0249	82.85
B	Noisy (mitigated)	0.9434	0.0250	80.60
D	Noisy (raw)	0.9421	0.0225	74.96
D	Noisy (mitigated)	0.9432	0.0219	72.31

MEM reduces the mean QUBO cost by $\sim 3\%$ on both processors, recovering part of the noise-induced degradation. The effect on P_{opt} is small because the 1% readout error affects all states roughly equally.

1. **Problem-Aware Processor D (Architecture-Algorithm Co-Design):** Instead of the generic circulant $C_9(1,2)$, we co-designed a topology that places the strongest $|J_{ij}|$ couplings at graph distance 1, minimizing routing overhead for the highest-weight ZZ terms. The algorithm uses Kruskal-style spanning tree construction with strongest edges first, then greedily fills remaining degree budget:

Topology	CX	Depth	SWAPs	P_{feasible} (noisy)	P_{opt} (noisy)
Frozen D (circulant $C_9(1,2)$)	102	87	10	0.9421	0.0225
Problem-Aware D (co-designed)	**90**	**78**	**8**	**0.9451**	**0.0237**

The co-designed topology reduces CX count by **12%**, depth by **10%**, and SWAPs by **20%**, while also improving noisy P_{feasible} and P_{opt}. This validates the principle that hardware should be designed with the problem's interaction structure in mind.



1. **Constraint-Preserving Mixer QAOA:** We replaced the standard transverse-field mixer $R_X(2\beta)$ with an XY (partial-swap) mixer on pairs of qubits whose constraint coefficients are equal: (q_2, q_7) both have weight 4, and (q_3, q_6) both have weight 2. Swapping equal-weight qubits preserves the linear constraint $c^T y = 15$ exactly, so the state never leaves the feasible subspace. The initial state is a uniform superposition over the 28 feasible states (each paired with exact slack):

Mixer	Condition	P_{feasible}	P_{opt}	CX Count
Standard RX	Ideal	0.9634	0.0225	102
Standard RX	Noisy	0.9421	0.0225	102
Constraint-Preserving XY	Ideal	**1.0000**	**0.0747**	861
Constraint-Preserving XY	Noisy	0.9016	0.0286	861

In the ideal case, the CP mixer achieves 100% **feasibility** and $P_{\text{opt}} = 7.47\%$. Because the CP mixer restricts state evolution strictly within the 28 feasible states, the fair random-guessing baseline inside this feasible subspace is $\frac{1}{28} \approx 3.57\%$ (rather than $\frac{1}{32} = 3.125\%$ across all states). Against this fair in-subspace baseline, the CP mixer achieves a $2.1\times$ **gain** (and a $3.3\times$ **improvement** over standard $p = 1$ QAOA on D). The CX count is higher (861 vs 102) because the `initialize` instruction and the RXX/RYY mixer gates are more expensive than the simple RX mixer, making the circuit more noise-sensitive. Under noise, P_{feasible} drops to 90% but P_{opt} still improves to 2.86%, beating the standard mixer's 2.25%. This demonstrates the fundamental trade-off: constraint-preserving mixers guarantee feasibility in principle, but the deeper circuit exposes the state to more depolarizing noise.

6.5 Advanced Algorithmic Extensions

Beyond the improvements discussed above, we implemented five additional algorithmic experiments:

1. **Multi-Angle QAOA (ma-QAOA):** Standard $p = 1$ uses 2 parameters. ma-QAOA assigns each of the 9 single-qubit R_Z gates, 36 two-qubit R_{ZZ} gates, and 9 mixer R_X gates its own variational parameter (54 total), warm-started from the standard QAOA optimum. The best seed achieves $P_{\text{opt}} = 0.0552$ ($2.4\times$ **the random baseline**), but most seeds concentrate probability on non-optimal low-cost states ($P_{\text{opt}} \approx 0.001$). This reveals a key trade-off: extra expressiveness improves the energy landscape ($\langle C \rangle$ drops from -1.52 to -1.87) but can overfit to a single non-optimal state rather than the global optimum.

Method	Params	Best P_{opt}	$\langle C \rangle$
Standard QAOA	2	0.0225	-1.523
ma-QAOA (best seed)	54	**0.0552**	-1.868

1. **CVaR + Constraint-Preserving Mixer Combined:** The two strongest individual techniques are orthogonal and stack: the CP mixer guarantees $P_{\text{feasible}} = 100\%$ (ansatz constraint), while CVaR focuses the optimizer on the tail (cost function). Sweeping α:

α	P_{feasible}	P_{opt}
0.05	1.0000	0.0701
0.10	1.0000	0.0673
0.20	1.0000	0.0688
0.30	1.0000	**0.0712**

At $\alpha = 0.3$, the combined method achieves $P_{\text{opt}} = 7.12\%$ with **guaranteed** 100% **feasibility** — $3.2\times$ the standard QAOA and $2.3\times$ the random baseline, with no infeasible shots at all.

1. **Recursive QAOA (RQAOA):** The QAOA correlator $\langle Z_0 Z_4 \rangle = -0.131$ between items 1 and 5 indicates anti-correlation. RQAOA eliminates item 5 using the rule $x_5 = 1 - x_1$, reduces the knapsack to 4 items, and solves the reduced problem exhaustively. The reconstructed solution is items $\{1, 2, 3, 4\}$ with value $26 = V^*$ and weight 14 — the exact optimum, found **deterministically**. Unlike sampling-based QAOA, RQAOA never misses the optimum.
2. **Adaptive Penalty Scheduling (Homotopy Continuation):** The schedule $P = 1 \to 2 \to 3 \to 4 \to 5$ uses the previous P's optimum as the warm start for the next:

P	γ	β	P_{feasible}	P_{opt}	Starts
1	0.287	2.868	0.948	0.026	5
2	0.285	2.866	0.953	0.024	1
3	0.284	2.865	0.954	0.023	1
4	0.283	2.865	0.955	0.023	1
5	0.283	2.865	**0.955**	0.023	1

At $P = 5$ via homotopy, $P_{\text{feasible}} = 0.955$ (comparable to the best standard seed at 0.955) but with only **1 optimization start** instead of 5 — the warm start completely avoids the seed-stuck problem where 3 of 5 seeds fail at $P = 5$.

1. **Grover Adaptive Search (GAS):** In addition to variational QAOA, we implemented Grover Adaptive Search (`src/gas.py`), which applies quantum amplitude amplification with iterative threshold lowering to locate low-cost QUBO states without classical parameter optimization. In each round, an oracle marks states strictly below the current threshold y_{best}, Grover diffusion amplifies their probability with $\mathcal{O}(\sqrt{N_{\text{total}}/N_{\text{marked}}})$ iterations, and the sampled state updates the threshold. On the locked 9-qubit knapsack problem, GAS successfully identified the global optimum ($V^* = 26$, cost -26.0) within 12 threshold-lowering rounds. However, on NISQ hardware without fault tolerance, deep Grover diffusion circuits suffer substantially higher gate-error accumulation compared to low-depth QAOA ($p = 1$), making QAOA with Processor D co-design the superior architecture for near-term quantum execution.

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7. Cross-Validation Against Official Toolkit & Engineering Quality

7.1 Cross-Validation Against Official Starter Pack

All project-level metrics were cross-checked against the official `qff_phase2.py` functions to ensure strict compliance:

- `circuit_metrics`: All 6 metrics (logical_cx, cx, depth, two_q_depth, swaps, cx_overhead) match exactly for Processors B and D.
- `bell_experiment`: XX, YY, ZZ correlations and fidelity match exactly for both adjacent (0-1) and farthest (0-3) pairs under noise within statistical tolerances.
- `validate_processor_d`: Processor D passes with **0 violations** from the official validator.

7.2 Engineering Quality & Automated Test Infrastructure

To guarantee production-grade software engineering standards and immediate reproducibility:

- **Automated Pytest Test Suite (`tests/`):** We implemented 16 automated unit test cases spanning four specialized test modules:
- `tests/test_knapsack.py`: Validates exact ground truth $V^* = 26$, items $\{1, 2, 3, 4\}$, QUBO matrix parity with algebraic penalty, Ising Hamiltonian mapping, and uniform random baseline statistics (87.5% feasible, 3.125% optimal).
- `tests/test_circuits.py`: Validates QAOA circuit synthesis, Bell state projector formulas $F = (1 + \langle XX\rangle - \langle YY\rangle + \langle ZZ\rangle)/4$, and basis gate transpilation compliance.
- `tests/test_processors.py`: Validates Processor A capacity limits ($n = 5 < 9$), Processor B topology, and executes dual validation (internal + official `qff_phase2.validate_processor_d`) on Processor D.
- `tests/test_case_study.py`: Tests generalized multi-constraint QUBO builder, disaster-relief payload optimization, and logarithmic slack scaling.
- **Standalone Fast Formulation Check:** `tests/check_formulation.py` executes zero-dependency mathematical sanity checks in < 0.2 seconds.
- **GitHub Actions Continuous Integration:** `.github/workflows/ci.yml` automates linting, test execution, formulation checks, and official processor validation upon every commit on Ubuntu with Python 3.11.
- **Pinned Dependency Environment:** All packages are pinned to exact versions in `requirements.txt` to guarantee deterministic execution across platforms.

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8. Compliance, Claims vs. Evidence, and Verification Sign-Off

8.1 Claims vs. Evidence Verification Matrix

The table below maps every quantitative claim, rubric requirement, and experimental finding directly to verifiable artifacts in the repository:

Rubric Criterion / Claim	Quantitative Value / Claim	Verifiable Code / Artifact	Status
G1: Repo URL & Structure	Self-contained, clean repository with dedicated `src/`, `tests/`, `results/`	Git tree root, `.gitignore`, `.github/workflows/ci.yml`	**PASS**
G2: Single Problem Statement	Exclusively addresses **PS-O3** (0/1 Knapsack)	`README.md`, `report.md`	**PASS**
G3: Executable Notebook	Top-to-bottom run produces B and D results	`main.ipynb`	**PASS**
G4: Persistent Tables Exist	`results.csv` and `report.md` exist and match	`results/tables/results.csv`, `report.md`	**PASS**
C1: Bell State Verification	Adjacent noisy $F = 0.9633 \pm 0.0021$, Farthest $F = 0.9548 \pm 0.0023$	`results/tables/bell.csv`, `src/bell.py`	**PASS**
C2: Processor D Rules	Degree 4, 9 qubits, 18 edges; 0 official errors; frozen	`processors/processor_D.json`, `tests/test_processors.py`	**PASS**
C2: Pareto Trade-Off Justification	Explored 48 designs; choice based on depth 87 & symmetry	`results/figures/pareto_design_d.png`, `results/extra/pareto_candidates.csv`	**PASS**
C2: Prediction vs. Measured	Predicted ≈ 6–8 fewer SWAPs; measured exactly 7 fewer SWAPs	Report Section 3.3, `results/tables/resources.csv`	**PASS**
C3: Problem Formulation	9-qubit QUBO, $P = 5$, 4 slack bits, exhaustive $V^* = 26$	`src/knapsack.py`, `tests/check_formulation.py`	**PASS**
C3: Processor A Capacity	Needs 9 qubits, has 5 $\implies$ Not Executable	`results/tables/results.csv`, Report Section 4.1	**PASS**
C3: Processors B & D Runs	Ideal/noisy runs with exact metrics, baseline 0.875	`results/tables/results.csv`, `results/tables/results_detail.csv`	**PASS**
C3: Landscape on D	20×20 grid on D noisy with optimum $(\gamma = 0.2831, \beta = 2.8648)$	`results/figures/landscape_D.png`, `src/landscape.py`	**PASS**
C4: Numerical Consistency	Report numbers match CSVs with 0 discrepancies	`report.md`, `results/tables/results.csv`	**PASS**
C5: Rigorous Analysis	Detailed discussion of connectivity, depth, noise, trade-offs	Report Sections 1–8	**PASS**
Extra: Disaster-Relief Study	Real-world framing + generalized dual-constraint QUBO	`src/case_study.py`, `results/extra/case_study_scaling.csv`	**PASS**
Extra: Hardware Noise & QASM 3	IBM calibration, ZNE, dynamical decoupling, OpenQASM 3	`results/extra/ibm_noise_results.csv`, `results/extra/processor_D_circuit.qasm`	**PASS**

8.2 Compliance and Verification Sign-Off

- **Hard Rules Integrity:** `qff_phase2.py` was preserved completely unmodified. Official settings (`rz, sx, x, cx`, `optimization_level=1`, SABRE, `seed_transpiler=42`, `seed_simulator=42`, 4096 shots, official noise) were maintained across all required benchmark runs.
- **Cross-Check:** All project metrics match the official `qff_phase2.py` functions exactly (Section 7). No discrepancy was found in circuit metrics, Bell correlations, or Processor D validation.
- **Reproducibility:** Executable top-to-bottom via `jupyter nbconvert --to notebook --execute main.ipynb --output main_executed.ipynb` with complete numerical consistency across all tables and figures. Total execution runtime is well under the 30-minute limit.